

Medikart

Pharmacy E-Commerce & Pharmacy Management System

Project Folder Structure

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This document lays out the full repository structure for Medikart, organized around how the project will actually be hosted: a Next.js storefront and an Express API running as managed processes on the VPS, and a React/Vite admin dashboard built to static files served by Nginx. The tree below is followed by a plain-language explanation of what each significant file and folder is for.

Folder Structure

Real `.env` files are shown because they genuinely exist on disk — the root `.gitignore` is what keeps them, `node_modules`, and build output out of Git. Only what's actually committed to the repository is meant to be tracked; everything else is either a secret, reproducible from a lockfile, or generated at build/deploy time.

```
medikart/
|-- apps/
|   |-- web/                                     # Customer storefront -- Next.js (SSR/SSG)
|       |-- app/
|           |-- layout.jsx
|           |-- page.jsx
|           |-- sitemap.js
|           |-- robots.js
|           |-- products/
|               |-- page.jsx
|               |-- [slug]/
|                   |-- page.jsx
|                   |-- opengraph-image.jsx
|           |-- categories/[slug]/page.jsx
|           |-- cart/page.jsx
|           |-- checkout/
|               |-- page.jsx
|               |-- confirmation/page.jsx
|           |-- instant-order/page.jsx
|           |-- about/page.jsx
|           |-- contact/page.jsx
|       |-- components/
|           |-- ui/
|               |-- LabelStub.jsx
|               |-- DiscountBadge.jsx
|               |-- Button.jsx
|               |-- PriceTag.jsx
|           |-- product/
|               |-- ProductCard.jsx
|               |-- ProductGrid.jsx
|               |-- ProductImage.jsx
|               |-- StockBadge.jsx
|           |-- cart/
|               |-- CartDrawer.jsx
|               |-- CartLineItem.jsx
|           |-- checkout/
|               |-- CheckoutStepper.jsx
|               |-- OtpInput.jsx
|               |-- DeliveryCityField.jsx
|               |-- PaymentMethodSelect.jsx
|               |-- PrescriptionUpload.jsx
|           |-- chatbot/
|               |-- ChatWidget.jsx
|               |-- ChatMessage.jsx
|           |-- whatsapp/
|               |-- WhatsAppButton.jsx
|       |-- layout/
|           |-- Header.jsx
|           |-- Footer.jsx
|   |-- lib/
|       |-- apiClient.js
|       |-- schema.js
|       |-- formatCurrency.js
|       |-- formatDiscount.js
|       |-- metadata.js
|   |-- public/
|       |-- favicon.ico
|       |-- logo.svg
```

```

| | | -- styles/
| | | | -- tokens.css
| | | | -- globals.css
| | | -- node_modules/           # installed, gitignored
| | | -- .env.local              # REAL secrets -- gitignored
| | | -- .env.local.example      # committed template
| | | -- next-sitemap.config.js
| | | -- next.config.js
| | | -- tailwind.config.js
| | | -- package.json
| | | `-- package-lock.json
|
| -- admin/                      # Staff dashboard -- React + Vite (SPA)
| | -- src/
| | | -- pages/
| | | | -- Overview/index.jsx
| | | | -- Products/
| | | | | -- ProductList.jsx
| | | | | -- ProductEdit.jsx
| | | | | -- NarcoticsView.jsx
| | | | | -- BulkImport.jsx
| | | | | -- MissingImages.jsx
| | | | -- Categories/index.jsx
| | | | -- Discounts/index.jsx
| | | | -- Orders/
| | | | | -- StandardOrders.jsx
| | | | | -- InstantOrders.jsx
| | | | | -- NarcoticsOrders.jsx
| | | | -- Cities/index.jsx
| | | | -- Reports/index.jsx
| | | | -- AdminUsers/
| | | | | -- UserList.jsx
| | | | | -- UserEdit.jsx          # role + module access (Super Admin only)
| | | | -- Settings/
| | | | | -- General.jsx
| | | | | -- Delivery.jsx
| | | | | -- Payment.jsx
| | | | | -- Email.jsx
| | | | | -- Prescription.jsx
| | | | | -- GoogleSheets.jsx
| | | | | -- Security.jsx
| | | | | -- Backup.jsx
| | | | | -- Pages.jsx            # About / Contact Us content
| | | | -- ActivityLogs/index.jsx
| | | | -- Login.jsx
| | | | -- ForgotPassword.jsx / ResetPassword.jsx
| | | -- components/
| | | | -- DataTable/
| | | | -- ProductForm/
| | | | | -- NarcoticsToggle.jsx
| | | | | -- DiscountField.jsx
| | | | -- OrderCard/
| | | | -- SearchBar.jsx
| | | | -- BulkActionBar.jsx
| | | | -- NotificationBell.jsx
| | | | -- RoleAccessEditor.jsx
| | | -- hooks/
| | | | -- useProducts.js
| | | | -- useProductSearch.js
| | | | -- useOrders.js
| | | | -- useAuth.js
| | | | -- useActivityLogs.js
| | | | -- useAdminUsers.js
| | | -- api/
| | | | -- products.js
| | | | -- orders.js
| | | | -- categories.js
| | | | -- cities.js

```

```

|-- discounts.js
|-- reports.js
|-- settings.js
`-- adminUsers.js

|-- store/
|-- authStore.js
|-- App.jsx
`-- main.jsx
-- node_modules/                # installed, gitignored
-- .env                          # REAL secrets -- gitignored
-- .env.example                  # committed template
-- index.html
-- vite.config.js
-- tailwind.config.js
-- package.json
`-- package-lock.json

-- server/                        # Backend API -- Node.js + Express
|-- src/
|-- modules/
|-- products/
|-- product.routes.js
|-- product.controller.js
|-- product.service.js          # incl. discount + narcotics-flag logic
|-- product.model.js
|-- product.validation.js
`-- productImages.controller.js # multi-image upload/delete/set-primary
-- categories/
|-- category.routes.js
|-- category.controller.js
|-- category.service.js
`-- category.model.js
-- cities/
|-- city.routes.js
|-- city.controller.js
|-- city.service.js
`-- city.model.js
-- discounts/
`-- discount.service.js          # shared precedence logic (product > category)
-- orders/
|-- order.routes.js
|-- order.controller.js
|-- order.service.js            # incl. cancellation: COD / void / refund
|-- order.model.js
|-- standardOrder.handler.js
|-- instantOrder.handler.js
|-- narcoticsOrder.handler.js
`-- cancelOrder.handler.js      # cancel order + auto void/refund
-- prescriptions/
|-- prescription.routes.js
|-- prescription.controller.js
`-- prescription.service.js
-- otp/
|-- otp.routes.js
|-- otp.controller.js
|-- otp.service.js
`-- otp.model.js
-- payments/
|-- payment.routes.js
|-- payment.controller.js
|-- payment.service.js          # incl. void()/refund() + charge()/authorize()
|-- providers/
|-- `-- habibMetro.provider.js
`-- `-- webhook.controller.js
-- admin-users/
|-- adminUser.routes.js
|-- adminUser.controller.js     # create/edit-role/delete -- Super Admin only
|-- adminUser.service.js

```

```

| | | | |-- adminUser.model.js
| | | | |-- passwordReset.controller.js
| | | | |-- passwordReset.service.js
| | | |-- chatbot/
| | | | |-- chatbot.routes.js
| | | | |-- chatbot.controller.js
| | | | |-- chatbot.service.js # Groq call + narcotics filter + disclaimer
| | | | |-- chatbotConversation.model.js
| | |-- activity-logs/
| | | |-- activityLog.routes.js
| | | |-- activityLog.service.js
| | | |-- activityLog.model.js
| | |-- reports/
| | | |-- report.routes.js
| | | |-- report.controller.js
| | | |-- report.service.js
| | |-- settings/
| | | |-- settings.routes.js
| | | |-- settings.controller.js
| | | |-- settings.service.js
| | | |-- settings.model.js # incl. discount + pages groups
|-- integrations/
| | |-- googleSheets.js
| | |-- smtp.js
| | |-- imageProcessor.js # sharp: resize/compress/WebP
| | |-- notifications.js # WebSocket/SSE to admin dashboard
| | |-- redisClient.js # cache + rate-limit + session/OTP store
| | |-- groqClient.js # chatbot LLM calls
|-- middleware/
| | |-- auth.js
| | |-- requireSuperAdmin.js
| | |-- errorHandler.js
| | |-- rateLimiter.js # Redis-backed, global default + per-route limits
| | |-- csrf.js
| | |-- validateRequest.js
|-- jobs/
| | |-- weeklyReport.js
| | |-- sheetSync.js
| | |-- narcoticsAuthExpiry.js
|-- config/
| | |-- db.js
| | |-- redis.js
| | |-- env.js
|-- app.js

|-- uploads/
| |-- products/
| | |-- {sku}/
| | | |-- original.jpg
| | | |-- thumb.webp
| | | |-- full.webp
| |-- prescriptions/ # PRIVATE -- authenticated route only
|-- placeholder.webp

|-- scripts/
| |-- importProducts.js
| |-- matchProductImages.js
| |-- seedNarcotics.js

|-- tests/
| |-- unit/
| | |-- discount.test.js
| | |-- deliveryCharge.test.js
| | |-- narcoticsGate.test.js
| | |-- otp.test.js
| | |-- cancelOrder.test.js # COD / void / refund branching (Phase 17)

|-- node_modules/ # installed, gitignored

```

```
| |-- .env                                # REAL secrets -- gitignored
| |-- .env.example                       # committed template
| |-- ecosystem.config.js                # PM2 process definitions
| |-- package.json
| |-- package-lock.json
|
|-- docs/
| |-- SRS.docx
| |-- PRD.docx
| |-- design.md
| |-- rules.md
| |-- phases.md
| |-- workflow-flowcharts.pdf
| |-- folder-structure.pdf
|
|-- nginx/
| |-- medikart.conf                      # reverse proxy + static routing (no secrets)
|
|-- .gitignore
|-- README.md
```

File & Folder Reference

What each significant file or folder is for, grouped the same way as the tree above.

apps/web — Customer Storefront (Next.js)

app/layout.jsx	Root layout shared by every page — loads fonts, design tokens, and global providers (cart context, etc.).
app/page.jsx	Home page — the search-first hero and featured categories.
app/sitemap.js	Generates sitemap.xml at build/request time from the live product and category data, for Google Search Console.
app/products/[slug]/page.jsx	A single product page, server-rendered so price, stock, and discount are visible to Google — not assembled client-side.
app/products/[slug]/open-graph-image.jsx	Auto-generates the preview image shown when a product link is shared on WhatsApp/social apps.
app/checkout/page.jsx	The Details -> OTP -> Payment -> Confirm checkout flow, including the narcotics prescription-upload gate.
app/instant-order/page.jsx	Prescription-only ordering flow — no product selection, per the Instant Order requirement.
app/about/page.jsx, app/contact/page.jsx	Static pages whose text content is pulled from Settings, so the client can edit wording without a developer.
components/ui/DiscountBadge.jsx	Renders the struck-through original price + discounted price + percentage-off badge.
components/checkout/PrescriptionUpload.jsx	Shared upload component used by both the Narcotics and Instant Order flows.
components/chatbot/ChatWidget.jsx	The floating AI assistant — sends symptom text to the backend chatbot endpoint and renders suggested products plus the mandatory disclaimer.
components/whatsapp/WhatsAppButton.jsx	The persistent floating icon linking to the client's WhatsApp Business number via a wa.me link.
lib/schema.js	Builds the JSON-LD structured data (Product, Organization, LocalBusiness) that powers rich search results.
lib/formatDiscount.js	Formats the original/discounted price pair and percentage-off text consistently everywhere it's shown.
.env.local / .env.local.example	Real secrets (API base URL, etc.) vs. the committed template teammates copy to create their own.

<code>next.config.js</code>	Next.js build/runtime configuration — image domains, redirects, etc.
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apps/admin — Staff Dashboard (React + Vite)

<code>pages/Products/NarcoticsView.jsx</code>	Filtered list of every product currently flagged Narcotics — the staff audit view, independent of the general catalog.
<code>pages/Discounts/index.jsx</code>	Where the storewide discount is set/cleared; product- and category-level discounts are set from their own edit screens.
<code>pages/Orders/NarcoticsOrders.jsx</code>	Where a pharmacist approves or rejects an uploaded prescription.
<code>pages/AdminUsers/UserEdit.jsx</code>	Super-Admin-only screen to set another admin's role and which modules they can access.
<code>pages/Settings/Pages.jsx</code>	Where the client edits the About and Contact Us page content shown on the storefront.
<code>pages/ForgotPassword.jsx / ResetPassword.jsx</code>	The admin password-recovery flow — request a reset link, then set a new password from it.
<code>components/ProductForm/NarcoticsToggle.jsx, DiscountField.jsx</code>	The specific form controls for flagging a product as Narcotics and setting its discount.
<code>components/SearchBar.jsx</code>	Product search-by-name/SKU on the Products list.
<code>components/RoleAccessEditor.jsx</code>	The module-level access checklist a Super Admin uses when editing another admin's permissions.
<code>api/*.js</code>	One file per backend module — every network call the dashboard makes is defined here, not scattered through components.
<code>.env / .env.example</code>	Real secrets (e.g. the API base URL) vs. the committed template.

server — Backend API (Node.js + Express)

<code>modules/**/*.routes.js</code>	Defines the URLs for that module and which controller function handles each one.
<code>modules/**/*.controller.js</code>	Reads the request, calls the service layer, shapes the response — no business logic lives here.
<code>modules/**/*.service.js</code>	The actual business rules — e.g. <code>product.service.js</code> decides how a product's discount and narcotics flag are applied.
<code>modules/**/*.model.js</code>	The MongoDB schema (Mongoose) for that module's data.
<code>modules/discounts/discount.service.js</code>	Shared logic implementing the confirmed precedence rule (product > category > storewide) so it's written once and reused everywhere a price is calculated.
<code>modules/orders/standardOrder.handler.js</code> , <code>instantOrder.handler.js</code> , <code>narcoticsOrder.handler.js</code>	The three distinct order-type workflows, kept separate rather than one tangled order function.
<code>modules/orders/cancelOrder.handler.js</code>	Cancels a Pending/Packed order (FR-AD-39) and decides the correct payment reversal — nothing to reverse for COD, a void for an authorized-only narcotics hold, or a real refund for an already-captured payment (FR-SYS-10) — plus the distinct cancellation email (FR-CW-24).
<code>modules/products/productImages.controller.js</code>	Handles multiple images per product (FR-AD-40): upload one or more files in a single request, mark one as the primary/cover image, delete a single image without touching the rest.
<code>modules/payments/payment.service.js</code>	The abstracted interface every provider implements: <code>charge()</code> , <code>authorize()</code> , and — since Order Cancellation — <code>void()</code> and <code>refund()</code> too, so <code>cancelOrder.handler.js</code> never needs to know which gateway is behind it.
<code>modules/payments/providers/habibMetro.provider.js</code>	The Habib Metro-specific integration code, isolated behind the shared <code>payment.service.js</code> interface so the gateway can be swapped without touching order logic.
<code>modules/payments/webhook.controller.js</code>	Receives and signature-verifies the gateway's payment-confirmation callback — the source of truth for whether an order was actually paid.
<code>modules/admin-users/adminUser.controller.js</code>	Create/edit-role/delete for other admin accounts — every route here requires the Super Admin check.
<code>modules/admin-users/passwordReset.*</code>	The forgot-password token flow, mirroring the OTP pattern already used at checkout.

<code>modules/chatbot/chatbot.service.js</code>	Calls Groq, filters out any Narcotics-flagged product from the suggestions, and attaches the mandatory disclaimer before the response is ever sent to a customer.
<code>modules/chatbot/chatbotConversation.model.js</code>	Logs each chatbot conversation for the periodic review this feature's compliance rule requires.
<code>integrations/redisClient.js</code>	Shared Redis connection used for catalog/price caching, rate-limit counters, and OTP/session storage.
<code>integrations/imageProcessor.js</code>	The sharp-based pipeline that resizes and converts every uploaded image to WebP before it's saved.
<code>integrations/groqClient.js</code>	Thin wrapper around the Groq API — kept separate from <code>chatbot.service.js</code> so the AI provider could be swapped later without touching the business rules.
<code>middleware/requireSuperAdmin.js</code>	Blocks any admin-account-management route unless the logged-in user is a Super Admin.
<code>middleware/rateLimiter.js</code>	Redis-backed token-bucket rate limiting applied as the default on every route — not only OTP/login/chatbot — with tighter overrides on those specific sensitive endpoints (Section 17.2 of the PRD).
<code>middleware/csrf.js</code>	Cross-site request forgery protection on admin state-changing requests.
<code>jobs/narcoticsAuthExpiry.js</code>	Scheduled job that auto-cancels a narcotics order's held card authorization if it's never approved within the configured window.
<code>uploads/products/{sku}/</code>	Where compressed product images actually live on disk, organized by SKU so they're easy to match back to a product.
<code>uploads/prescriptions/</code>	Prescription images — deliberately outside any public Nginx route, reachable only through an authenticated admin endpoint.
<code>scripts/importProducts.js</code>	One-time/repeatable bulk Excel import of the client's product catalog, with per-row error reporting.
<code>scripts/matchProductImages.js</code>	Matches the bulk AI-generated image folder to the correct product by SKU.
<code>scripts/seedNarcotics.js</code>	Applies the client's manually-provided Narcotics product list, cross-checked against the keyword-scan candidate list.
<code>tests/unit/</code>	Unit tests for the business logic where a silent bug would cost money or create a compliance gap: discount precedence, delivery-charge calculation, the narcotics checkout gate, OTP verification, and order cancellation's COD/void/refund branching.

<code>ecosystem.config.js</code>	The PM2 file that defines and keeps alive the two long-running Node processes on the VPS: the Express API and the Next.js server.
<code>.env</code> / <code>.env.example</code>	Real secrets (Mongo URI, Redis URL, SMTP, Habib Metro keys, Groq API key, JWT secret) vs. the committed template.

Root & Deployment Files

<code>nginx/medikart.conf</code>	Routes traffic: the storefront domain to the Next.js PM2 process, /api to the Express PM2 process, the admin subdomain to the static Vite build, and /uploads to the images folder. Safe to commit — contains no secrets.
<code>.gitignore</code>	Keeps <code>.env</code> files, <code>node_modules</code> , build output, and uploaded images out of Git — see the note below.
<code>README.md</code>	Setup instructions: environment variables needed, how to run each app locally, and the deploy steps for the VPS.
<code>docs/</code>	Every planning document produced so far — SRS, PRD, design direction, engineering rules, phases, and this folder structure — versioned alongside the code.

Why `.env` is real but never committed: `.env` holds actual secrets — the live MongoDB connection string, Redis URL, SMTP credentials, and eventually the Habib Metro gateway keys. If it were committed, anyone with repository access would have live credentials to the database and payment gateway. `.env.example` is the safe, committed template — same variable names, no real values — so anyone setting up the project knows exactly what needs to be filled in without ever seeing the actual secrets.

How This Maps to Hosting

Three things run on the VPS, coordinated by Nginx and PM2 exactly as already planned:

<code>apps/web</code>	Built and run as a Node process (<code>`next start`</code>), kept alive by PM2 via <code>ecosystem.config.js</code> . Nginx reverse-proxies the storefront domain to it.
<code>apps/admin</code>	Built once (<code>`vite build`</code>) into static files; Nginx serves them directly — no running process needed for this one.
<code>server</code>	Runs as a second Node process under the same PM2 file. Nginx reverse-proxies /api requests to it, and serves /uploads directly from <code>server/uploads/</code> as static files.

Redis and MongoDB Atlas sit alongside this: Redis runs locally on the VPS (or a small managed instance) since the API talks to it constantly and low latency matters; MongoDB stays on Atlas as already decided.